

NAGALAND LITERACY AND NUMERACY FEST 2026

Micro-Improvement Project (Numeracy Teacher) Grade 7

Chapter 7 — Working with Fractions

Why Micro-Improvement Project (MIP)?

Across Nagaland, fractions already live inside daily life: a woman at the Kohima market weighing out half a kilo of pork or a quarter kilo of axone, a weaver dividing thread for a mekhala, a family sharing a plate of Galho equally among guests, or a group of students dividing a stall's rice among hundreds of visitors at the Hornbill Festival. This Micro-Improvement Project (MIP) helps students notice these everyday acts of sharing and scaling, name them mathematically, and move from splitting a shared fruit by eye toward confidently multiplying and dividing fractions using rules they discover for themselves. It builds directly on students' existing sense of a fraction as “a part of a whole” from earlier grades, and prepares them to reason with fractions as operators and quantities rather than only as shapes on a page.

Skill in Focus	Working with Fractions — Multiplication and Division
Duration	1 Week
Competency Addressed	C-1.6: “Explores and applies fractions (in ratio and decimal form) in daily-life situations.”

Learning Outcome

The Grade 7 syllabus offers two learning-outcome tracks for this cycle — Integers, and Fractions & Rational Numbers. This MIP selects the most suitable two outcomes from the Fractions & Rational Numbers track, chosen because they directly build the operations students need for the recipe-scaling and fair-sharing tasks below, with word problems woven through every task rather than taught as a separate final step:

- Multiplication of fractions, including fraction as an operator (finding “a fraction of” a quantity).
- Reciprocal of a fraction and division of fractions, applied to word problems involving mixed fractions.

Key Concepts

- Demonstrates multiplication of a fraction by a whole number and by another fraction.
- Explains fraction as an operator — finding “a fraction of” a given quantity.
- Deduces the reciprocal of a fraction and the rule for dividing fractions, through pattern and generalisation.
- Provides counter-examples to common misconceptions about fraction multiplication and division.
- Solves word problems involving multiplication and division of fractions, including mixed fractions.

Detailed Tasks

Task-1

Sharing the Orange — Fractions as Parts of a Whole

Objective	Refresh students' intuitive understanding of a fraction as an equal part of one whole, using a real, shareable object, before moving on to operations on fractions.
Setting	Indoor (or outdoor under a tree, if preferred) — Classroom
Teacher's Note	This is a short recap task. Bring one real orange, tangerine, roti, or rice-cake per group if possible; a paper circle works just as well where fruit is not available. Keep the tone playful — this sets the stage for the whole week.
Duration	35 minutes
Materials Needed	1 orange / tangerine / roti / rice-cake per group, or a paper circle drawn on scrap paper · chalk or marker · notebook and pencil

Activity Steps

1. Show the class one orange (or a drawn circle). Ask: 'If 4 friends want to share this orange equally, how much does each friend get?' Take 2–3 responses.
2. In groups, students cut or draw the orange/circle into 4 equal parts. Each part is labelled $\frac{1}{4}$.
3. Repeat with different group sizes (2, 3, and 6 friends) and, where possible, a different item each time, recording the fraction each person receives in their notebook.

4. Introduce the vocabulary formally: numerator (the number of parts taken) and denominator (the total number of equal parts).
5. Close with a discussion: 'If we share the same orange among more and more friends, does each person's share get bigger or smaller? Why?'

Concept Built: *A fraction as an equal part of one whole; the meaning of numerator and denominator.*

Task-2

Scaling the Recipe — Multiplying a Fraction by a Whole Number

Objective	Help students multiply a fraction by a whole number, connecting it to repeated addition, using a recipe-scaling context that all students recognise.
Setting	Indoor (Classroom)
Teacher's Note	Write a simple 1-plate recipe on the board before class (for example, rice = $\frac{1}{2}$ cup, vegetables = $\frac{1}{4}$ cup per plate). Keep the numbers small and the denominators easy (halves, quarters, thirds) so the pattern is easy to see.
Duration	30 minutes
Materials Needed	A teacher-made recipe card or the same written on the board · small cups, bottle caps, or bowls to act as 'measuring cups' · notebook and pencil

Activity Steps

1. Present the 1-plate recipe on the board. Ask: 'How much rice do we need for 3 plates?' Let students first solve it by adding $\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$.
2. Show the shortcut on the board: $\frac{1}{2} \times 3 = \frac{3}{2} = 1 \text{ whole and } \frac{1}{2}$. Explain the rule step by step — multiply the numerator by the whole number, keep the denominator the same, then simplify or convert to a mixed number.
3. Group Task: each group scales a second ingredient from the recipe for 5 plates, and then for 8 plates, recording their working in a simple table (Ingredient | For 1 plate | For 5 plates | For 8 plates).
4. Differentiation Tip: for learners who struggle, use bottle caps as physical 'half-cup' units and have them count out caps one plate at a time before writing the multiplication.

Concept Built: *Multiplying a fraction by a whole number; converting an improper fraction to a mixed number.*

Task-3

Fraction of a Fraction — Multiplying Two Fractions

Objective	Introduce multiplication of a fraction by another fraction using a paper-folding (area) model, and build the meaning of 'fraction as an operator' — finding a fraction of a given quantity.
Setting	Indoor (Classroom)
Teacher's Note	This task uses paper folding, which every student can do at their own desk without any special material. If a woven Naga shawl, mat, or a picture of one is available, use it to open the discussion, but it is optional — the folding activity works on its own.
Duration	35 minutes
Materials Needed	1 rectangular strip of paper per student (old newspaper, exercise-book page, or scrap paper) · sketch pens or pencil · grid paper, if available · a Naga shawl or woven mat picture (optional)

Activity Steps

1. Ask: 'If only half of a shawl panel is woven, and only half of that woven half is dyed, what fraction of the whole shawl is dyed?' Let students guess before folding.
2. Students fold their paper strip in half and shade one half ($\frac{1}{2}$). They fold it in half again along the same direction and shade half of the already-shaded part, revealing that this section is $\frac{1}{4}$ of the whole strip — showing $\frac{1}{2}$ of $\frac{1}{2} = \frac{1}{4}$.
3. Repeat with a strip folded into thirds, then halved, so students see $\frac{1}{2}$ of $\frac{1}{3} = \frac{1}{6}$ by counting the small sections.
4. Guide students toward the rule: multiply the numerators together, multiply the denominators together, then simplify to the lowest form.
5. Introduce the operator language explicitly: 'Find $\frac{2}{3}$ of 12 mangoes' is solved as $\frac{2}{3} \times 12$. Demonstrate on the board.

6. Group Task: each group solves three 'find a fraction of' word problems using familiar local items (mangoes, betel nuts, plates of food, metres of cloth).

Concept Built: *Multiplying a fraction by another fraction using a folding/area model; fraction as an operator; simplifying to the lowest form.*

Task-4

Busting the Myths — Does Multiplying Always Make Numbers Bigger?

Objective	Confront and correct the common misconception that multiplication always makes a number bigger, and that a bigger fraction times a bigger fraction always gives the biggest answer, using worked counter-examples.
Setting	Indoor (Classroom)
Teacher's Note	This task works best right after Task-3, while the folding model is still fresh. Keep a running 'Myth or Fact?' chart on the classroom wall for the rest of the week.
Duration	30 minutes
Materials Needed	Grid or graph paper · chart paper for the class 'Myth or Fact?' board · sketch pens

Activity Steps

1. Write two multiplications on the board: $\frac{1}{2} \times 6$ and $\frac{1}{2} \times \frac{1}{2}$. Ask students to predict which answer will be bigger than $\frac{1}{2}$, and which will be smaller, before calculating.
2. Calculate together: $\frac{1}{2} \times 6 = 3$ (bigger than $\frac{1}{2}$), but $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ (smaller than $\frac{1}{2}$). Students discover that multiplying by a fraction less than one shrinks the number, while multiplying by a whole number greater than one grows it.
3. Present a second pair: $\frac{3}{4} \times \frac{2}{3}$ and $\frac{3}{4} \times 4$. Ask students to calculate both and compare which operation gives a smaller result and why.
4. In groups, students write one 'Myth' (a common wrong belief, e.g. 'Multiplying two fractions always gives a smaller fraction than both of them' or 'Multiplying always makes a number bigger') and one 'Fact' (their correct counter-example with numbers), and pin it to the class 'Myth or Fact?' chart.

5. Close by asking each group to explain their counter-example aloud to the class in their own words.

Concept Built: *Multiplying by a fraction less than one produces a smaller result; same-looking operations do not always behave the way we expect — demonstrated through worked counter-examples.*

Task-5

The Flip Trick — Cracking the Code for Dividing Fractions

Objective	Guide students to discover the reciprocal of a fraction and the rule for dividing fractions through pattern recognition and physical sharing, rather than being told the rule directly.
Setting	Indoor (Classroom)
Teacher's Note	Use small stones, beans, or seeds so students can physically act out sharing before the rule is generalised. Build the pattern table together on the board as students report their results.
Duration	35 minutes
Materials Needed	Small stones, dried beans, or seeds (about 20–30 per group, as counters) · grid paper or notebook for the pattern table · ruler

Activity Steps

1. Pose a sharing problem: 'We have $\frac{1}{2}$ kg of rice to share equally among 4 people. How much does each person get?' Let groups use stones to represent the half-kilo and physically split it into 4 equal piles.
2. Students record their result ($\frac{1}{2} \div 4 = \frac{1}{8}$ kg) in a three-column table: Dividend, Divisor, Quotient. Repeat with two or three more examples such as $\frac{1}{2} \div 2$, $\frac{3}{4} \div 2$, and $\frac{3}{4} \div 3$, physically sharing counters each time.
3. Ask students to look at the table and find a pattern connecting the divisor to the quotient, without repeating the physical sharing every time. Guide them toward noticing that dividing by 4 gives the same answer as multiplying by $\frac{1}{4}$.
4. Formalise the rule: to divide by a number, multiply by its reciprocal — flip the numerator and denominator (a whole number n becomes the fraction $\frac{1}{n}$).

5. Students verify the rule using a real example: $\frac{3}{4}$ kg of bamboo shoot is shared evenly across 6 plates. Demonstrate: $\frac{3}{4} \div 6 = \frac{3}{4} \times \frac{1}{6} = \frac{3}{24} = \frac{1}{8}$ kg per plate.

Concept Built: *The reciprocal of a fraction, and the rule for dividing fractions — dividing by a number is the same as multiplying by its reciprocal — deduced through pattern and generalisation.*

Task-6

Measure My Kitchen

Objective	Apply multiplication and division of fractions to a real recipe or sharing situation at home, connecting classroom learning to daily life in students' own households.
Setting	Homework — done at home with family, discussed in class the next day
Teacher's Note	Explain this clearly one day in advance. Encourage students to pick a dish their family actually cooks, not an imaginary one — the numbers do not need to be exact, only realistic.
Duration	20 minutes
Materials Needed	A measuring cup, spoon, or any container used at home · notebook and pencil

Activity Steps

1. Each student chooses one real dish cooked at home and, with a family member's help, notes the fractional quantity of one main ingredient used for the usual number of people served (e.g. ' $\frac{1}{2}$ cup of rice for 2 people').
2. Using multiplication of fractions, students calculate how much of that ingredient would be needed if the family had double or triple the usual number of guests.
3. Using division of fractions, students then calculate how much one person's fair share would be if that same total amount had to be shared among a different number of people (for example, fewer plates than usual, so each share is a fraction of the total).
4. Students write one sentence explaining what the multiplication told them (e.g. how much extra ingredient is needed for more guests) and one sentence explaining what the division told them (e.g. each person's fair share).
5. The next day, students share their chosen dish and calculations with a partner or the class.

Concept Built: *Transfers multiplication and division of fractions to a self-chosen, real-life sharing situation, strengthening the link between classroom mathematics and daily life.*

Culminating Showcase

Showcase

Nagaland Feast Planners — Final Project

Objective	Bring together every concept from Tasks 1–5 into one creative, real-world design project that groups plan, calculate, and present.
Setting	Indoor (combined classroom project) — 2–3 sessions after Task-6
Teacher's Note	Spread this across 2–3 sessions after Task-6, culminating in a short class exhibition. Encourage groups to be as creative as possible with their chosen feast or stall context.
Duration	30–35 minutes per session (2–3 sessions)
Materials Needed	Chart paper or cardboard · ruler, pencil, crayons or sketch pens · small stones, beans, or bottle caps as counters · locally available materials for models (bamboo sticks, leaves, small stones) — optional

Activity Steps

1. In groups of 4–5, students choose a feast or sharing occasion meaningful in their community — a village feast, a church community lunch, a school sports-day meal, or a Hornbill Festival food stall.
2. Groups decide on one dish and a chosen number of guests or plates (for example, 20, 30, or 50), and use multiplication of fractions to scale their recipe from 1 plate up to that number, showing their working clearly on the chart.
3. Groups then imagine a different, smaller number of actual guests arriving and use division of fractions to work out each guest's fair share, so that nothing is wasted.
4. Groups identify and label at least one place in their design where they deliberately used the idea from Task-4 — for instance, showing that multiplying by a fraction made an ingredient amount smaller, not bigger, and explaining why this matters for their plan.

5. In the final session, each group presents its design in a short class exhibition, explaining the dish, the scaling and sharing calculations, and one interesting or surprising discovery from their design process.

Concept Built: *Independent application of fraction multiplication, division, the reciprocal rule, and misconception-awareness in one integrated, creative design task.*

BY THE END OF THIS PROJECT, TEACHERS ALONG WITH STUDENTS ARE REQUIRED TO COME UP WITH INNOVATIVE PROJECTS WHICH CAN BE EXHIBITED WITHIN THE CLASSROOM.

By the end of these tasks, teachers and students should have moved from splitting a shared orange to confidently multiplying, dividing, and applying fractions — culminating in an original, locally-grounded feast or stall plan that each group can showcase with pride.

Reflect on Students' Understanding

Observe 5–6 students each week during the tasks above. No written test is needed — watch what students do with counters, folded paper, and real recipes, and listen to how they explain their thinking.

Student Name	Explains Fraction as Part	Multiplies Fractions	Divides Fractions (Reciprocal)	Corrects a Misconception	Needs Support
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

Scale (for each column): ☐ Not Yet ☐ Sometimes ☐ Consistently

Picture Guide

Photocopy or redraw this one-page summary as a poster for the classroom wall.



WORKING WITH FRACTIONS



NAGALAND - FRACTIONS IN DAILY LIFE

WHAT IS A FRACTION?



$$= \frac{1}{4}$$

A fraction is a part of a whole.

KEY IDEAS

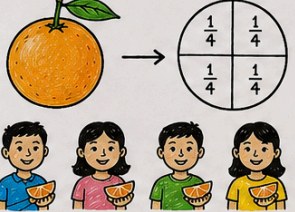
- Part of a whole
- Multiply fractions
- Divide using reciprocal
- Use in daily life.



LEARNING GOALS

- Multiply a fraction by a whole number
- Multiply one fraction by another
- Divide fractions
- Solve problems

1. SHARING THE ORANGE



One whole = $\frac{4}{4}$

2. RECIPE SCALING

Recipe for 1 plate

Rice = $\frac{1}{2}$ cup

Veg = $\frac{1}{4}$ cup

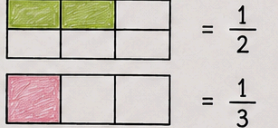
For 3 plates:

Rice = $\frac{1}{2} \times 3 = \frac{3}{2} = 1 \frac{1}{2}$ cup

Veg = $\frac{1}{4} \times 3 = \frac{3}{4}$ cup

Multiply the numerator.
Keep the denominator.

3. FRACTION OF A FRACTION



Take $\frac{1}{3}$ of $\frac{1}{2}$

$\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

Multiply across:
Numerator x Numerator
Denominator x Denominator

4. DIVIDING FRACTIONS

$\frac{2}{3} \div \frac{3}{4} = ?$

Step 1: Change $\div$ to $\times$

$\frac{2}{3} \times \frac{4}{3}$

Step 2: Multiply

$\frac{2 \times 4}{3 \times 3} = \frac{8}{9}$

To divide, multiply by the reciprocal.

IN OUR DAILY LIFE


 $\frac{1}{2}$ kg pork


 $\frac{1}{3}$ of shawl


 $\frac{1}{4}$ of Galho


Sharing food at festival

OUR TAKEAWAY

- ★ Fractions help us share.
- ★ We can multiply and divide.
- ★ Fractions make life easier!

